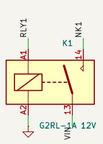
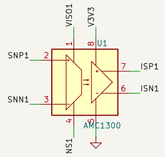
PHASE 1 — relay-isolated boost leg (0 deg carrier)

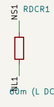




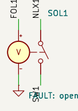




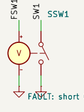


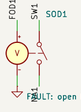


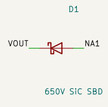




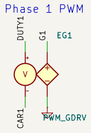






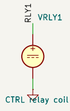


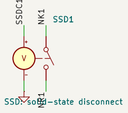
Phase 1 PWM + gate drive, controller I/O and fault injection







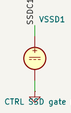




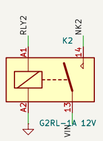




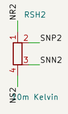


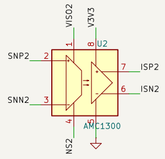


PHASE 2 — relay-isolated boost leg (180 deg carrier)





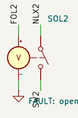




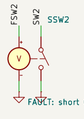


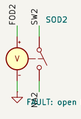






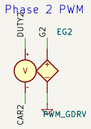






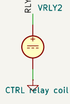


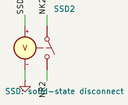
Phase 2 PWM + gate drive, controller I/O and fault injection

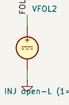




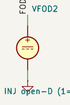


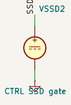












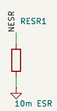
INPUT BUS, OUTPUT BUS, FEEDBACK, AUX SUPPLY





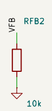




























Controller I/O (VDUTYk, VRLYk, VSSDk) default to open-loop values so the sheet runs in the KiCad simulator as-is.
sim/run_case.py rewrites them as ngspice 'external' sources driven by the Python firmware model
(fault detection, debounce, template correlation, relay isolation, closed-loop reconfiguration)
and rewrites VFOLk / VFSWk / VFODk as PWL steps at t = 50 ms for the fault-injection matrix.

.tran 50n 80m 0 50n

Generated by sim/gen_schematic.py — edit the generator, not this file
2 x 1 mH, 50 kHz, 180 deg interleaved; per-phase isolated sensing + relay isolation

Sheet: /
File: ft_boost.kicad_sch

Title: Fault-tolerant redundant-phase interleaved boost (48 V -> 200 V)

Size: A2	Date: 2026-10-18	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1